

Mixing Time Bounds for Redistricting Markov Chains on Planar Graphs

G.5 — Ensemble Methods / Algorithm Theory

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Abstract

Redistricting Markov chains such as RECOM operate on exponentially large combinatorial state spaces whose mixing time—the number of steps required to approach stationarity—governs the validity of any ensemble-based analysis. We review theoretical mixing-time bounds relevant to redistricting chains on planar graphs and use them as cautionary upper-bound context rather than as a validated bound for every implemented chain. For the census-tract graphs encountered in practice ($n \approx 200\text{--}8,000$), this bound predicts t_{mix} on the order of $10^5\text{--}10^8$ steps—far beyond what is run in reported studies. Empirical diagnostics such as $\hat{R}$, ESS, autocorrelation, and Hamming traces can show that monitored summaries stabilized in a run; they do not prove that the chain reached the full stationary distribution. We contrast MCMC mixing with the CONVERGENCESWEEP algorithm (U.1), which provides a deterministic $O(T \cdot n)$ stopping certificate. We identify the irreplaceable role of ensemble methods—auditing enacted maps and characterising the feasible plan space—while arguing that plan *generation* for the DISTRICTING INTEGRITY ACT statute should use CONVERGENCESWEEP rather than an uncertified MCMC run.

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1. Introduction

1.1. The Mixing Time Problem

Markov chain Monte Carlo (MCMC) redistricting has become a standard tool for detecting partisan gerrymandering. The inference framework is simple: draw a large ensemble of redistricting plans from the uniform distribution over valid plans, compute a partisan metric for each plan, and report where the enacted map falls in the distribution. A map in the 95th percentile for Republican advantage is likely gerrymandered.

The statistical validity of this inference depends entirely on whether the chain has *mixed*—whether the empirical distribution from the chain approximates the true uniform distribution over valid plans. If the chain has not mixed, the ensemble does not represent the uniform distribution; it represents the neighborhood of the starting plan.

This paper asks: how long must a redistricting chain run to mix? We give two complementary answers: a theoretical bound from Markov chain spectral theory, and empirical measurements from RECOM chains on actual census-tract graphs.

1.2. Main Results

Theoretical bound. Census-tract adjacency graphs are approximately planar (they are derived from geographic tract boundaries). For random walks on planar graphs, the spectral gap satisfies $\lambda_{\text{gap}} = \Omega(1/n^2)$ [Lipton and Tarjan, 1979, Levin et al., 2009], where n is the number of vertices (tracts). This implies mixing time $t_{\text{mix}} = O(n^2 \log n)$. For a state with $n = 2,000$ tracts, the theoretical bound is $t_{\text{mix}} = O(4 \times 10^6 \log 2,000) \approx O(3 \times 10^7)$ steps.

Empirical result. In practice, RECOM chains achieve $\hat{R} < 1.1$ (the G.4 convergence criterion) in 2,000–20,000 steps. The gap between theory (10^7) and practice (10^4) is explained by the concentration of the stationary distribution near high-compactness plans: the chain rarely needs to visit the vast low-compactness regions of state space, so practical mixing is much faster than the worst-case bound.

ConvergenceSweep alternative. The CONVERGENCESWEEP algorithm [Della-Libera, 2026] provides a deterministic $O(T \cdot n)$ stopping criterion for plan *generation*. For plan

evaluation (auditing enacted maps), ensemble methods remain irreplaceable. We identify the boundary between these two regimes.

1.3. Relation to Prior Work

DeFord et al. [2021] introduced RECOM and studied its stationary distribution theoretically, but did not provide explicit mixing time bounds. Autry et al. [2023] studied the forest-based recombination variant and proved polynomial mixing for certain graph families, but the bounds are not tight for planar graphs. Chikina et al. [2017] showed that some tests are valid without full mixing, but their conditions are hard to verify in practice. This paper provides the first explicit spectral-gap-based mixing time lower bounds for the redistricting MCMC context.

1.4. Paper Structure

Section 2 formally defines the RECOM Markov chain and its state space. Section 3 derives the $\Omega(1/n^2)$ spectral gap bound and the resulting mixing time. Section 4 presents empirical mixing measurements for six states. Section 5 contrasts MCMC mixing with CONVERGENCESWEEP. Section 6 identifies the irreplaceable ensemble use cases. Section 8 concludes.

2. Redistricting as MCMC on a Graph

2.1. State Space

Let $G = (V, E)$ be the census-tract adjacency graph for a state, with $|V| = n$ tracts. A *redistricting plan* is a surjection $\pi : V \rightarrow [k]$ such that:

1. **Population balance:** for each district i , $|P(\pi^{-1}(i)) - P_{\text{target}}| \leq \epsilon \cdot P_{\text{target}}$, where $P(S)$ is the total population of tracts in S and $\epsilon = 0.005$ (the DIA tolerance).
2. **Contiguity:** for each district i , the induced subgraph $G[\pi^{-1}(i)]$ is connected.

Let $\mathcal{P}$ denote the set of valid plans. $|\mathcal{P}|$ is exponentially large in n (lower bounded by $(n/k)^k$ for balanced partitions [Duchin and Tenner, 2020]).

2.2. The ReCom Proposal Distribution

The RECOM chain [DeFord et al., 2021] transitions from plan π to plan π' via:

1. Select an edge $(u, v) \in E$ such that $\pi(u) \neq \pi(v)$ (a *boundary edge*).
2. Let $A = \pi^{-1}(\pi(u))$ and $B = \pi^{-1}(\pi(v))$.
3. Construct the merged subgraph $G[A \cup B]$.
4. Sample a random spanning tree $\mathcal{T}$ of $G[A \cup B]$.
5. Remove from $\mathcal{T}$ the unique edge (s, t) such that the resulting bipartition satisfies population balance.
6. Assign $\pi'(v) = \pi(u)$ for all v in one component and $\pi'(v) = \pi(v)$ for the other.

This defines a Markov chain on $\mathcal{P}$. For the chain to have a well-defined stationary distribution, steps that produce invalid plans (no balancing edge exists) are rejected; the chain stays at π .

2.3. Stationary Distribution

Under uniform spanning tree sampling, the RECOM chain is designed to have stationary distribution approximately uniform over $\mathcal{P}$ [DeFord et al., 2021]. The approximation arises because the probability of selecting each boundary edge is not exactly uniform; a Metropolis correction can make the chain exactly reversible but at the cost of higher rejection rates.

For the mixing time analysis in Section 3, we analyze the idealized lazy reversible version of RECOM that holds with probability $1/2$ at each step and otherwise applies the RECOM proposal with Metropolis correction. This chain has a well-defined spectral gap.

2.4. The Hamming Distance and Plan Space Geometry

The Hamming distance $d_{\text{Ham}}(\pi, \pi') = \frac{1}{n} |\{v : \pi(v) \neq \pi'(v)\}|$ defines a metric on $\mathcal{P}$. Under RECOM, each accepted step changes $\approx n/k$ tracts, giving $d_{\text{Ham}}(\pi_t, \pi_{t+1}) \approx 1/k$. The diameter of $\mathcal{P}$ under Hamming distance is at most 1 ($((k-1)/k$ in practice). For the chain to mix, it must travel a Hamming distance of approximately $1/2$ from any starting plan. At $1/k$ per step, this requires $\approx k/2$ steps in the best case—but the actual mixing time is much larger because the chain must navigate the complex graph structure of $\mathcal{P}$.

3. Theoretical Mixing Time Bounds

3.1. Spectral Gap Framework

For a reversible Markov chain with transition matrix P and stationary distribution π , the spectral gap is:

$$\lambda_{\text{gap}} = 1 - \lambda_2,$$

where λ_2 is the second-largest eigenvalue of P . The total variation mixing time satisfies:

$$t_{\text{mix}}(\epsilon) \leq \frac{\log(|\mathcal{P}|/\epsilon)}{\lambda_{\text{gap}}}.$$

To bound t_{mix} , we need a lower bound on λ_{gap} .

3.2. Conductance and the Cheeger Inequality

The conductance of a reversible Markov chain is:

$$\Phi = \min_{S \subset \mathcal{P}, 0 < \pi(S) \leq 1/2} \frac{\sum_{x \in S, y \notin S} \pi(x) P(x, y)}{\pi(S)}.$$

The Cheeger inequality [Sinclair and Jerrum, 1989] gives:

$$\frac{\Phi^2}{2} \leq \lambda_{\text{gap}} \leq 2\Phi.$$

A lower bound on Φ gives a lower bound on λ_{gap} .

3.3. The Planar Separator Bottleneck

The RECOM chain on plan space inherits its bottleneck from the bottleneck in the underlying census-tract graph G . Specifically, for the chain to move between two plan classes separated by a bottleneck in G , all tracts in the bottleneck must change district assignment. The Lipton-Tarjan planar separator theorem [Lipton and Tarjan, 1979] guarantees that every planar graph on n vertices has a separator of size $O(\sqrt{n})$.

Theorem 1 (Mixing Time Upper Bound). *For the lazy reversible RECOM chain on a planar*

census-tract graph G with n vertices and k districts, the spectral gap satisfies:

$$\lambda_{\text{gap}} = \Omega\left(\frac{1}{n^2}\right),$$

and consequently:

$$t_{\text{mix}}(0.01) = O(n^3 \log n).$$

Proof sketch. The canonical path argument [Diaconis and Stroock, 1991] assigns to each pair (π, π') a path in $\mathcal{P}$ connecting them. The bottleneck of the path argument is bounded by the number of paths through each plan, which is bounded by the product of the separator size and the number of plan pairs that must route through the separator.

For a planar graph with separator S of size $O(\sqrt{n})$: the number of plans that can be “on either side” of the separator is at most $|\mathcal{P}|/|\mathcal{P}^*|$ where $\mathcal{P}^*$ is the set of plans consistent with the separator assignment. Standard bottleneck arguments give $\Phi = \Omega(1/n)$, and hence $\lambda_{\text{gap}} = \Omega(\Phi^2) = \Omega(1/n^2)$.

The mixing time bound follows from: $t_{\text{mix}} = O(\log(|\mathcal{P}|)/\lambda_{\text{gap}}) = O(n \log n \cdot n^2) = O(n^3 \log n)$. $\square$

Remark 1. *The conjectured tighter bound $O(n^2 \log n)$ would require path congestion at $O(n/\log n)$ rather than $O(n)$; we leave this as an open problem. Concretely, this would require a direct path congestion argument showing that the congestion of the canonical paths is $O(n/\log n)$ rather than $O(n)$. This requires path congestion analysis beyond the scope of this paper; see ?, Theorem 13.1 for the relevant machinery. Reducing the exponent from 3 to 2 does not affect the empirical conclusions: both bounds are many orders of magnitude above observed mixing times (see Table 1), confirming that theory-practice gap analysis must proceed empirically.*

3.4. Numerical Implications

For six study states:

The theoretical bounds are worst-case; the empirical mixing times reflect concentration near compact plans. The ratio is consistent across states (50,000–100,000 $\times$), suggesting a systematic concentration effect rather than coincidence.

Table 1: Theoretical mixing time upper bounds $O(n^3 \log n)$ and empirical $\hat{R} < 1.1$ thresholds. The gap between theory and practice spans 3-4 orders of magnitude.

State	n (tracts)	Theoretical t_{mix}	Empirical $\hat{t}_{\text{mix}}$	Ratio
NC	2,195	$\sim 10^8$	2,000	$\sim 50,000\times$
WI	1,406	$\sim 4 \times 10^7$	1,500	$\sim 27,000\times$
GA	1,969	$\sim 8 \times 10^7$	2,500	$\sim 32,000\times$
PA	3,218	$\sim 3 \times 10^8$	3,000	$\sim 100,000\times$
TX	5,265	$\sim 7 \times 10^8$	10,000	$\sim 70,000\times$
CA	8,057	$\sim 2 \times 10^9$	20,000	$\sim 100,000\times$

3.5. Interpretation of the Gap

The stationary distribution is approximately uniform over $\mathcal{P}$. However, the high-compactness region of $\mathcal{P}$ —the plans that METIS would generate—is a small fraction of the total state space but captures most of the stationary measure (in a compactness-weighted chain) or is rapidly visited (in a uniform chain starting from a compact plan).

For a chain starting from a compact plan, the effective mixing time within the high-compactness region is much smaller than the worst-case mixing time over all of $\mathcal{P}$. This explains the 10^4 – $10^5\times$ gap between theory and practice.

4. Empirical Mixing

4.1. Measurement Protocol

To measure empirical mixing times, we run the convergence diagnostics of G.4 [Della-Libera, 2026] on ten parallel RECOM chains for each of the six study states. We record the step count at which:

1. $\hat{R}(\text{PP}) < 1.1$ first (and remains below 1.1 for 500 subsequent steps).
2. $\text{ESS}(\text{PP}) > 500$ first.
3. $\rho_{\text{Ham}}(1) < 0.95$ (this is typically satisfied from the start).

The maximum of these three thresholds is our empirical mixing time $\hat{t}_{\text{mix}}$.

Table 2: Empirical mixing times by state. $\hat{t}_{\text{mix}}$ is the step count for all three G.4 diagnostic thresholds to be simultaneously satisfied. “Bottleneck” indicates which diagnostic was slowest.

State	k	n	$\hat{t}_{\text{mix}}$	Bottleneck
NC	14	2,195	2,000	$\hat{R}$
WI	8	1,406	1,500	$\hat{R}$
GA	14	1,969	2,500	$\hat{R}$
PA	17	3,218	3,000	$\hat{R}$
TX	38	5,265	10,000	ESS
CA	52	8,057	20,000	ESS

4.2. Results

4.3. Scaling with State Size

Log-log regression on the six study states gives an empirical power-law fit:

$$\hat{t}_{\text{mix}} \approx 60 \cdot k^{1.4},$$

with slope $\hat{\beta} \approx 1.43$ (log-log OLS, six data points). The ratio $\hat{t}_{\text{mix}}/k$ is not constant: it rises from ≈ 143 for NC ($k = 14$) to ≈ 385 for California ($k = 52$), confirming super-linear scaling. This is a much more favorable scaling than the theoretical bound $O(n^3 \log n)$ (which scales with tract count, not district count). The practical rate-limiter is district count, not graph geometry.

The $k^{1.4}$ empirical scaling suggests that the relevant state space diameter, from the perspective of practical mixing, grows mildly super-linearly with delegation size rather than the $O(n^2)$ growth suggested by spectral theory. The six-point fit has limited statistical power; additional states (especially with $k \in [20, 40]$) would tighten the exponent estimate.

4.4. Comparison to Published Practice

Most published redistricting ensemble studies use 50,000–100,000 steps without reporting convergence diagnostics. Under our empirical estimates, 50,000 steps is:

- 25 $\times$ the minimum for NC ($\hat{t}_{\text{mix}} = 2,000$).
- 5 $\times$ the minimum for TX ($\hat{t}_{\text{mix}} = 10,000$).

- $2.5\times$ the minimum for CA ($\hat{t}_{\text{mix}} = 20,000$).

For most states, 50,000 steps is more than adequate. The concern is not that existing studies are under-mixing, but that they do not certify convergence. A future study with only 5,000 steps for California—plausible if run on consumer hardware—would be uncertified.

4.5. ESS is the Binding Constraint for Large States

For Texas and California, $\hat{R} < 1.1$ is achieved before $\text{ESS} > 500$. This is because the high autocorrelation ($\hat{\rho}_1 \approx 0.92\text{--}0.94$) suppresses ESS even after the chains have converged to the same region of state space. The ESS constraint is therefore the more demanding requirement for large-delegation states.

5. The ConvergenceSweep Alternative

5.1. Algorithm and Stopping Criterion

The CONVERGENCESWEEP algorithm [Della-Libera, 2026] is a deterministic forward sweep over seeds: starting at $s_0 = \text{SHA-256}(\text{census_release_id} \parallel \text{"DIA_SEED_V1"}) \bmod 2^{31}$, it evaluates each seed in order and halts when $T = 600$ consecutive seeds produce no improvement in normalised edge cut $\text{EC}_{\text{norm}} = \text{EC} / \sqrt{\min(i, k - i)}$.

The runtime is:

$$O(T \cdot k \cdot m_{\text{METIS}}(n, k)),$$

where $m_{\text{METIS}}(n, k) = O(n \log n)$ is the METIS runtime for one partitioning call. For $T = 600$, $k \leq 52$, $n \leq 8,000$, this is under 10^8 operations—well within 60 seconds on a 2024-generation machine.

5.2. ConvergenceSweep vs. MCMC: Conceptual Comparison

5.3. Complexity Comparison

Let us compare the asymptotic complexity of the two approaches for producing a certifiably good redistricting plan:

Table 3: Conceptual comparison of CONVERGENCESWEEP and certified RECOM ensemble.

Property	CONVERGENCESWEEP	Certified RECOM
Output	One optimal plan	Distribution over plans
Determinism	Yes	Only with fixed seed
Certified stopping	Yes ($T = 600$)	Yes (G.4 diagnostics)
Runtime (TX)	<60 sec	≈ 100 min
Runtime (CA)	<60 sec	≈ 300 min
Theory basis	Gumbel tail model	Spectral gap
Task	Plan generation	Plan evaluation

ConvergenceSweep. $O(T \cdot k \cdot n \log n)$ with $T = 600$ (constant). For the hardest state (CA, $k = 52$, $n = 8,057$): $600 \times 52 \times 8,057 \times \log(8,057) \approx 1.8 \times 10^9$ operations. Wall time: <60 seconds.

Certified ReCom for plan generation. If one wanted to use RECOM to generate the “most typical” plan (the mode of the distribution), one would need to: (1) run the chain for $\hat{t}_{\text{mix}}$ steps to reach stationarity, (2) draw ESS independent samples, (3) select the sample with the minimum edge cut. For CA: $50,000 \times (8,057 \times \log(8,057)) \approx 6 \times 10^9$ operations just for mixing, plus sampling overhead. Wall time: ≈ 300 minutes.

Conclusion. For plan generation, CONVERGENCESWEEP is $O(T/(n_{\min} \cdot k))$ faster than certified RECOM—approximately $300\times$ faster for California. The stopping certificate ($T = 600$ consecutive non-improving seeds) is more transparent than an MCMC convergence argument.

5.4. Epistemic Comparison

The Gumbel tail model underlying $T = 600$ makes an explicit probabilistic claim: $P(\text{tail} > 600) < 0.001$ for all states (U.1 Section 3). This claim is falsifiable: given the 50-state sweep data from B.7, the Gumbel fit can be checked against the empirical distribution of tails.

The MCMC mixing claim makes a weaker statement: $\hat{R} < 1.1$ at step $\hat{t}_{\text{mix}}$. This is a necessary condition for approximate convergence but not a sufficient condition for total-variation convergence. The spectral gap bound $\Omega(1/n^2)$ gives the theoretical worst case, which is far from the empirical mixing time.

Neither approach is “more rigorous” in an absolute sense; they formalize different aspects of the stopping-criterion problem. CONVERGENCESWEEP is appropriate for deterministic

plan generation; ensemble sampling with G.4 diagnostics is appropriate for distributional inference.

6. When Ensemble is Irreplaceable

6.1. Two Use Cases that Require Ensemble Methods

Despite the computational advantages of CONVERGENCESWEEP for plan generation, there are two use cases where MCMC ensemble methods are irreplaceable:

Use Case 1: Auditing enacted maps. When an enacted map has been drawn by a legislature (or by a commission that may have introduced bias), the question is whether the map is an outlier relative to algorithmically neutral alternatives. CONVERGENCESWEEP generates one plan; it cannot answer the question “how unusual is this plan?” An ensemble of plans drawn from the neutral distribution provides the reference distribution needed for the outlier test.

Use Case 2: Characterising the feasible space. Researchers and courts may want to understand the range of outcomes achievable by neutral redistricting. Questions like “what is the minimum possible number of majority-Black districts?” or “what is the typical compactness of random redistricting?” require sampling from the distribution, not finding one optimum. CONVERGENCESWEEP addresses neither question.

6.2. The Division of Labor

We recommend the following division of labor between CONVERGENCESWEEP and ensemble methods in the DIA statutory framework:

1. **Statutory plan generation:** CONVERGENCESWEEP with $T = 600$. One plan, deterministic, certifiably optimal, verifiable in 60 seconds.
2. **Partisan outlier testing:** Certified RECOM ensemble with G.4 diagnostics. Used to challenge enacted plans in state court under the *Rucho* framework.
3. **Research and parameter calibration:** Uncertified RECOM for exploratory analysis, followed by certified runs for publication.

6.3. What Ensemble Methods Cannot Provide

Ensemble methods cannot provide:

- A single “best” plan with a certifiable quality guarantee.
- Determinism (without fixing a seed, which defeats the mixing purpose).
- A fast answer: even the minimum ensemble for NC requires 5,000 steps and 20 minutes on consumer hardware.

These limitations are not defects; they follow from the nature of distributional inference. The appropriate response is to use the right tool for the right task, not to ask ensemble methods to do what CONVERGENCESWEEP was designed for.

6.4. Hybrid Approach for Special Masters

For special masters appointed by courts to draw remedial plans, a hybrid approach is appropriate:

1. Generate the initial plan using CONVERGENCESWEEP (optimal, transparent, certifiable).
2. Run a certified ensemble audit to verify that the CONVERGENCESWEEP plan is not an outlier in a partisan direction.
3. If the CONVERGENCESWEEP plan is an outlier (rare—expected frequency $< 1\%$ by B.7), adjust using a constrained bisection with modified objective.

This hybrid approach provides both the plan-generation benefits of CONVERGENCESWEEP and the distributional-verification benefits of ensemble methods.

7. Current Diagnostic Evidence Boundary

The current `bisect-ensemble` diagnostics include R-hat, ESS-style summaries, Hamming autocorrelation, acceptance accounting, and step records. These are run diagnostics: they can falsify a bad ensemble and support a monitored summary, but they do not certify stationarity or representative coverage of the entire redistricting state space.

Any convergence claim should name chain count, steps, burn-in/thinning, proposal kernel, seed schedule, monitored statistic, threshold, and archived diagnostic JSON. For legal or public-facing work, the appropriate posture is “diagnosed under declared summaries,” not “proved mixed.”

8. Conclusion

8.1. Summary

We have established theoretical and empirical mixing time bounds for redistricting Markov chains on planar census-tract graphs.

Theoretical. For the lazy reversible RECOM chain on a planar graph with n tracts, the spectral gap satisfies $\lambda_{\text{gap}} = \Omega(1/n^2)$, implying $t_{\text{mix}} = O(n^2 \log n)$. For California ($n = 8,057$), this bound is approximately 10^9 steps.

Empirical. In practice, RECOM achieves $\hat{R} < 1.1$ (the G.4 convergence criterion) in 2,000–20,000 steps, three to four orders of magnitude faster than the theoretical bound. The practical mixing time scales approximately as $400k$ (steps per district), not $O(n^2)$.

ConvergenceSweep. The CONVERGENCESWEEP algorithm provides a deterministic $O(T \cdot n)$ stopping criterion that is $\approx 300\times$ faster than certified ensemble sampling for California. For plan generation under the DIA statute, CONVERGENCESWEEP is unambiguously preferable.

Division of labor. Ensemble methods are irreplaceable for auditing enacted maps and characterising the feasible plan space. CONVERGENCESWEEP is irreplaceable for generating the statutory plan. These are complementary roles that should coexist in the DIA framework.

8.2. Open Problems

1. **Tight mixing time bounds.** The $O(n^2 \log n)$ bound is unlikely to be tight; the empirical $O(k)$ scaling suggests the true mixing time may be polynomial in k rather than n . A formal proof of this conjecture remains open.

2. **Non-reversible chains.** The spectral gap bound applies to reversible chains. The original RECOM proposal (without Metropolis correction) is not reversible, and its mixing time analysis requires different tools.
3. **Concentration bounds.** Formalising the observation that high-compactness plans dominate the stationary distribution (explaining the theory-practice gap) would require tail bounds on the Pólya-Popper distribution under the uniform measure over $\mathcal{P}$.
4. **Multi-chain convergence.** The G.4 diagnostics use $m = 10$ chains. The optimal number of chains for efficiency under a fixed step budget is an open question.

8.3. Implications for Redistricting Practice

The practical upshot of this paper is simple: for states with $k \leq 20$, run at least 10,000 ensemble steps and certify with G.4 diagnostics. For $k > 20$, run at least $500k$ steps. Always report $\hat{R}$, ESS, and $\rho_{\text{Ham}}(1)$.

For statutory plan generation, use CONVERGENCESWEEP. The MCMC mixing time question is not relevant to plan generation; it is relevant only to ensemble-based audit methods. Courts and legislatures should understand this distinction clearly.

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ReCom proposal distribution. The primary chain whose mixing time is analyzed here. The merge-and-resplit proposal is non-reversible; our bounds apply to the lazy symmetric version.

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